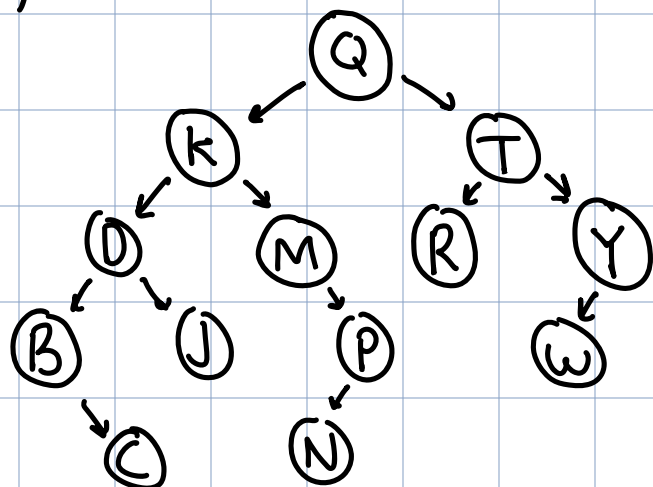
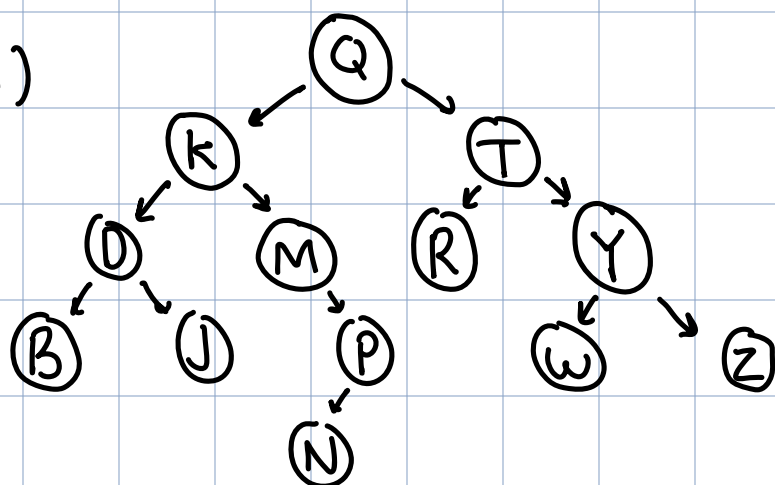


Brent Foxworth - Homework 2

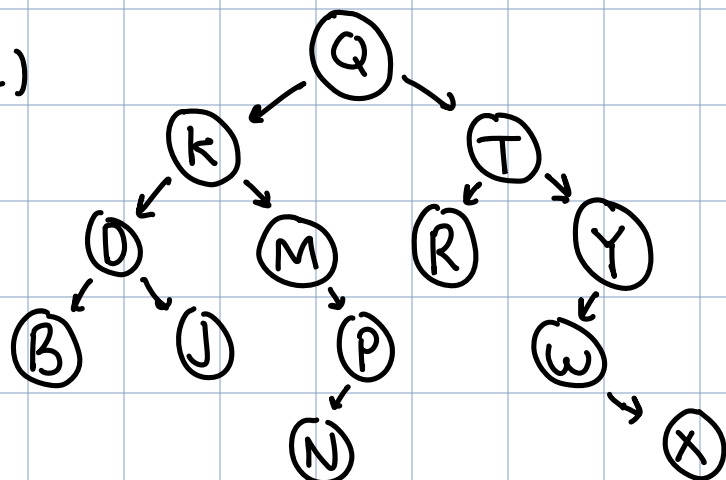
8a)



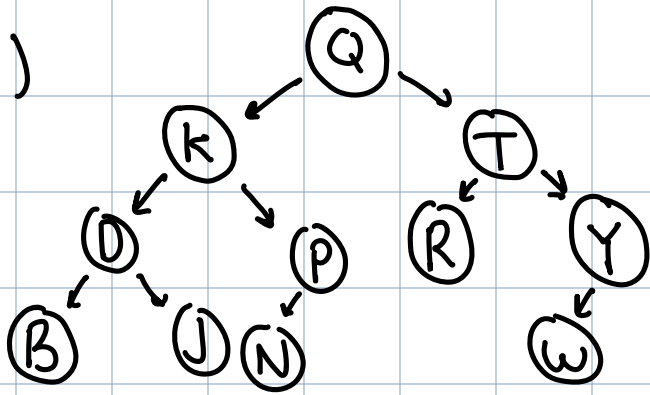
8b)



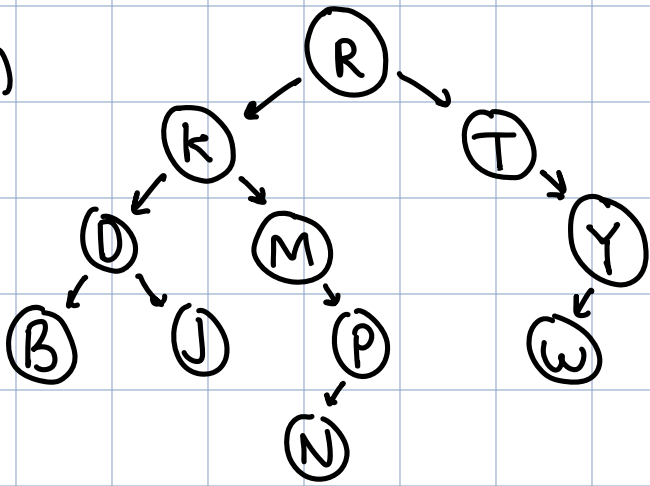
8c)



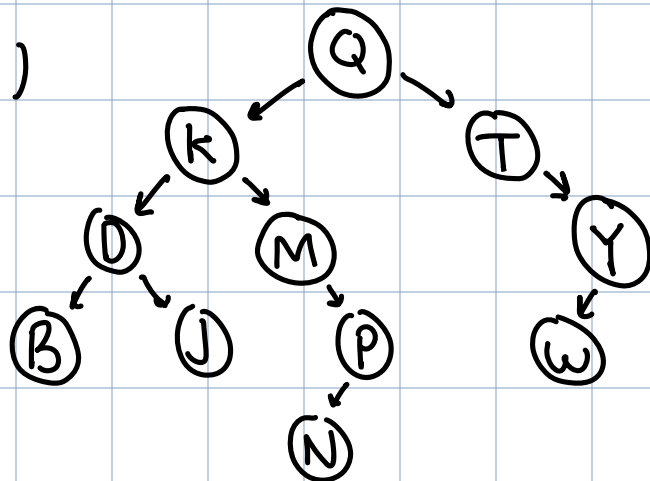
8d)



8e)



8f)

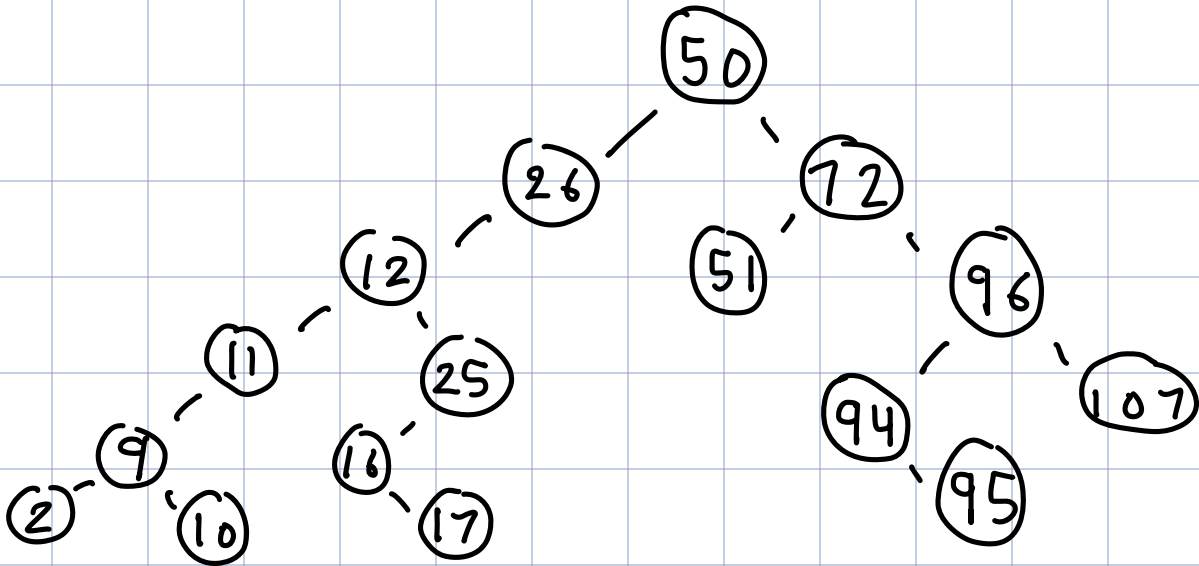


9a) B, D, J, K, M, N, P, Q, R, T, W, Y

9b) B, J, D, N, P, M, K, R, W, Y, T, Q

9c) Q, K, D, B, J, M, P, N, T, R, Y, W

10.

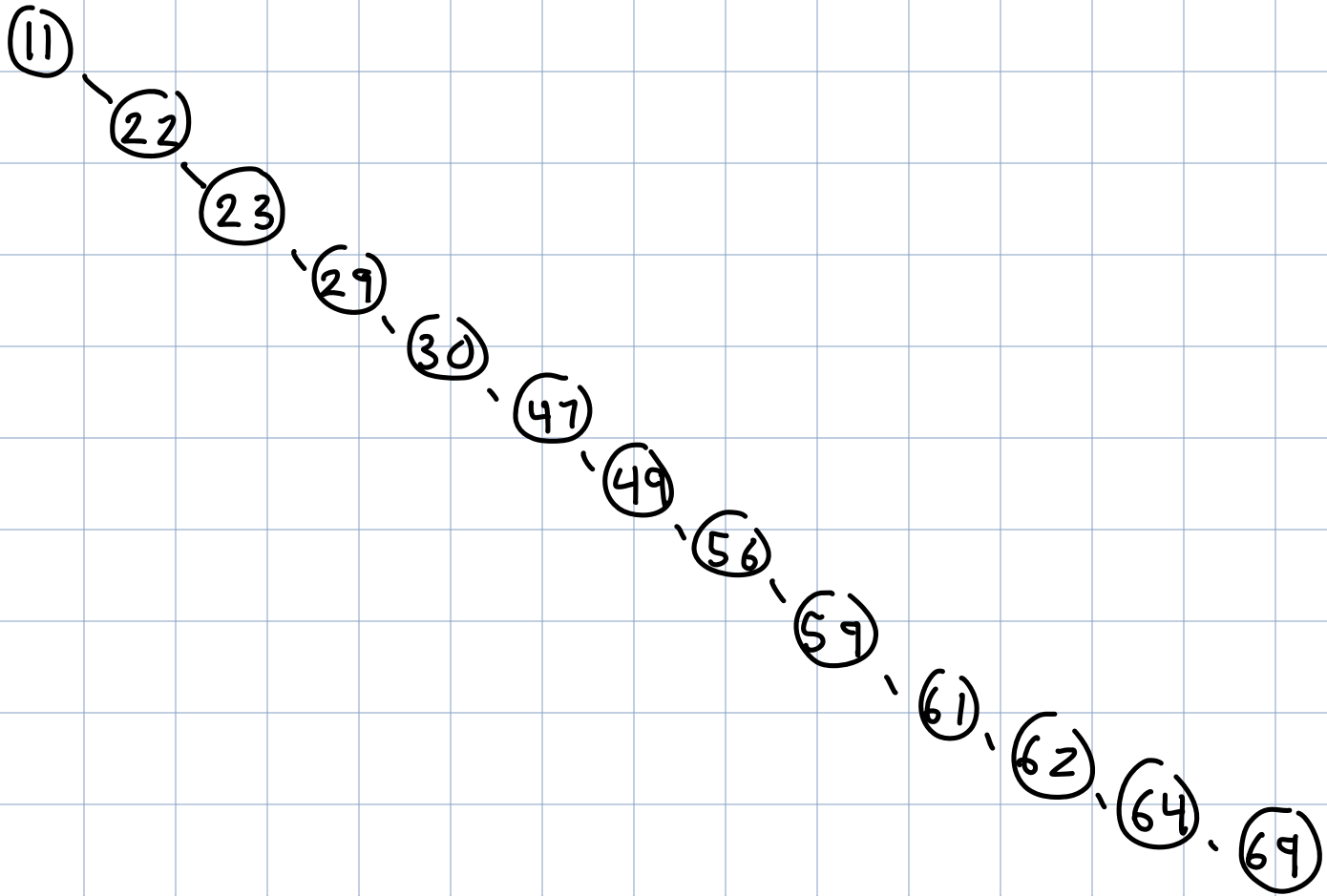


11 a) 5

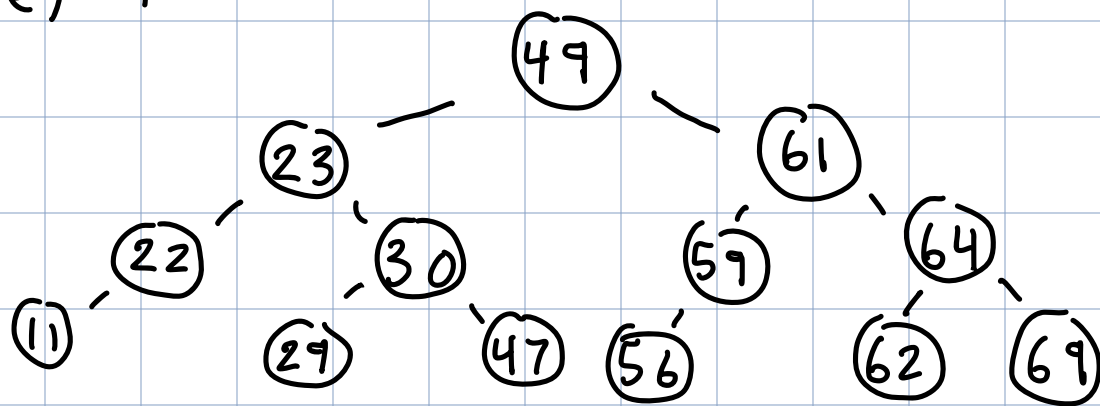
11 b) 22, 49, 59

11 c) 1, 2,

11 d) 13



11e) 4



12a) 56, 69, 59, 62, 61

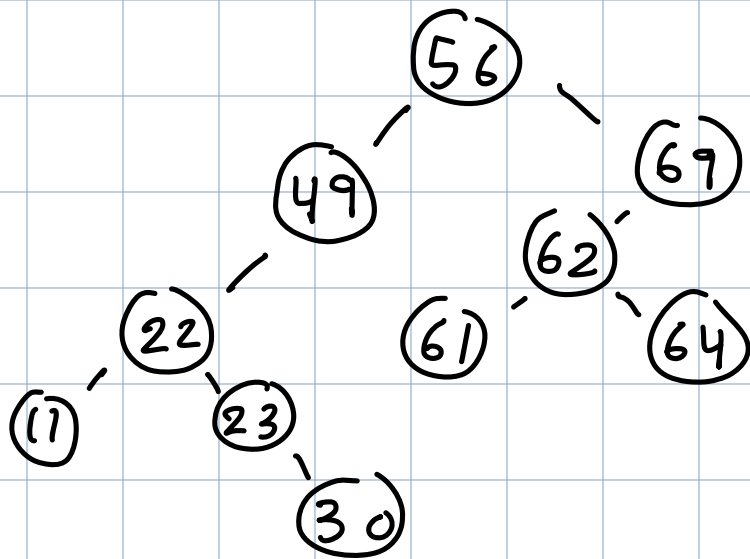
12b) 56, 47, 22, 29, 23

13a) 11, 22, 23, 29, 30, 47, 49, 56, 59, 61, 62, 64, 69

13b) 11, 23, 30, 29, 22, 49, 47, 61, 64, 62, 59, 69, 56

13c) 56, 47, 22, 11, 29, 23, 30, 49, 69, 59, 62, 61, 64

14.



15.

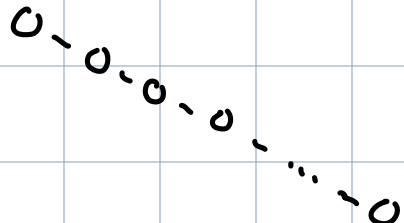
```
graph TD; 56((56)) --- 47((47)); 56 --- 69((69)); 47 --- 22((22)); 47 --- 49((49)); 22 --- 11((11)); 22 --- 29((29)); 11 --- 9((9)); 11 --- 10((10)); 29 --- 23((23)); 29 --- 30((30)); 49 --- 48((48)); 69 --- 59((59)); 59 --- 62((62)); 62 --- 61((61)); 62 --- 64((64)); 64 --- 63((63)); 64 --- 77((77)); 77 --- 76((76));
```

16a) False

16b) False

16 c) False

16 d) False

17. NO. The in order traversal of a binary search tree will result in the nodes being put in order from least to greatest. When rebuilding the binary search tree it would result in a skewed tree more like a linked list: 

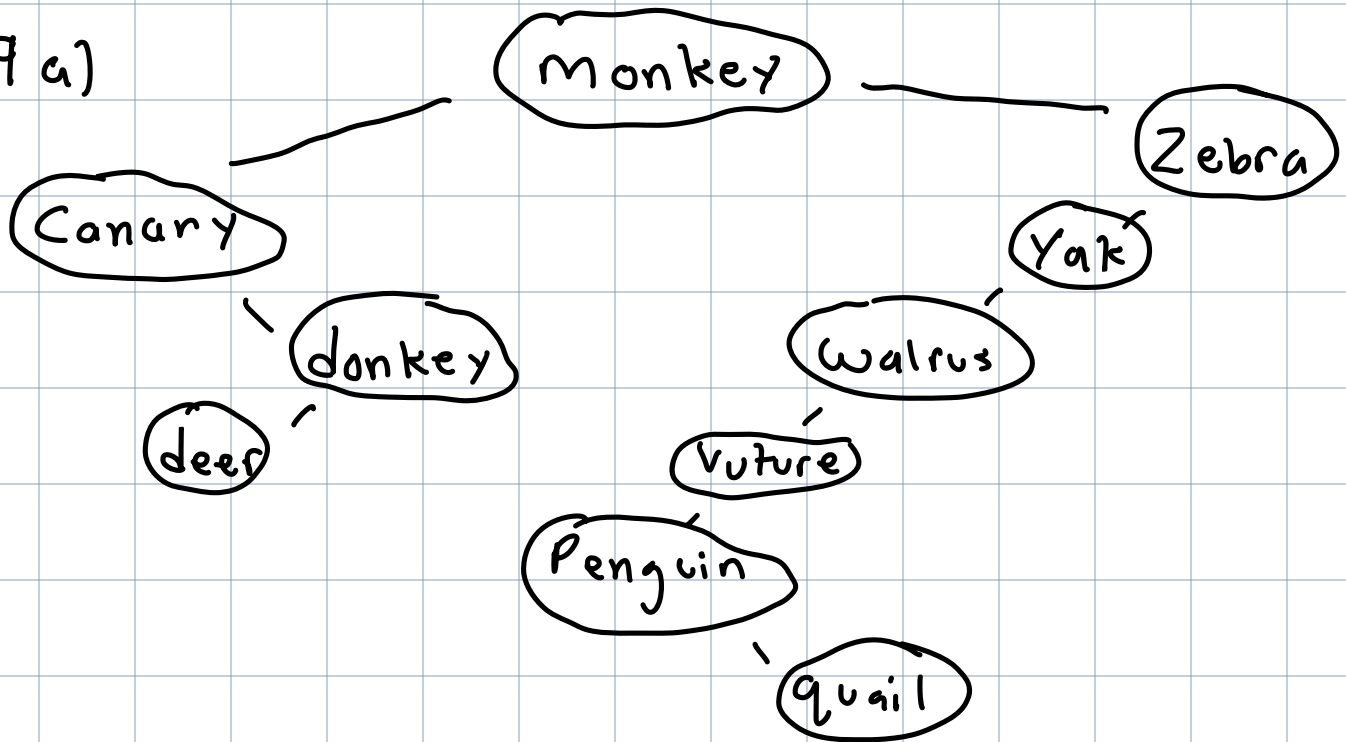
18 a)

	Best	worst	average
Order linked list	$O(1)$	$O(n)$	$O(n)$
Binary Search tree	$O(1)$	$O(n)$	$O(\log n)$

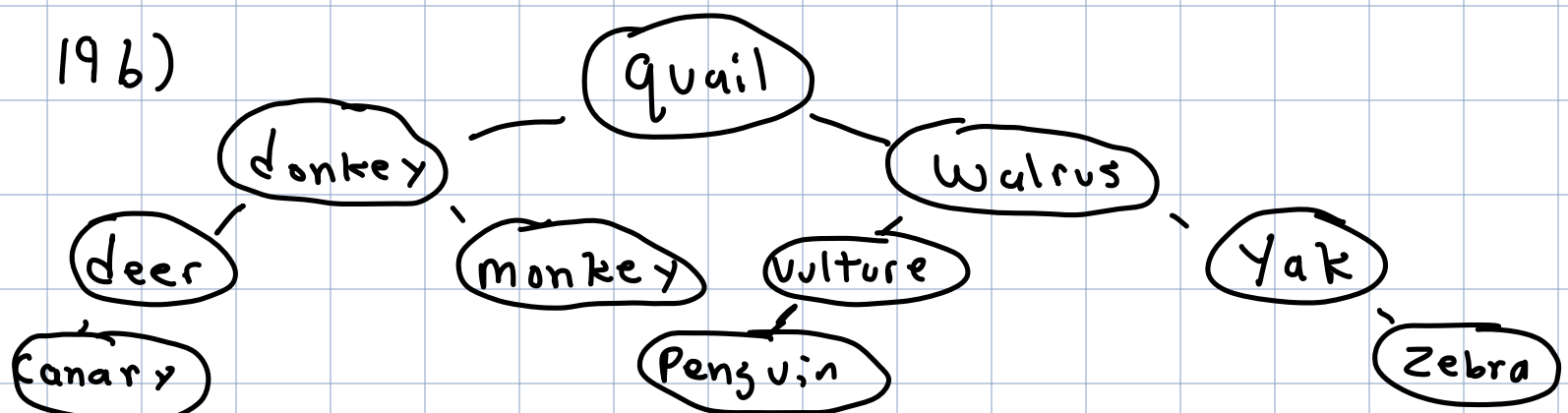
18 b)

	Best	worst	average
Order linked list	$O(1)$	$O(n)$	$O(n)$
Binary Search tree	$O(1)$	$O(n)$	$O(n)$

19 a)



19 b)



19c)

